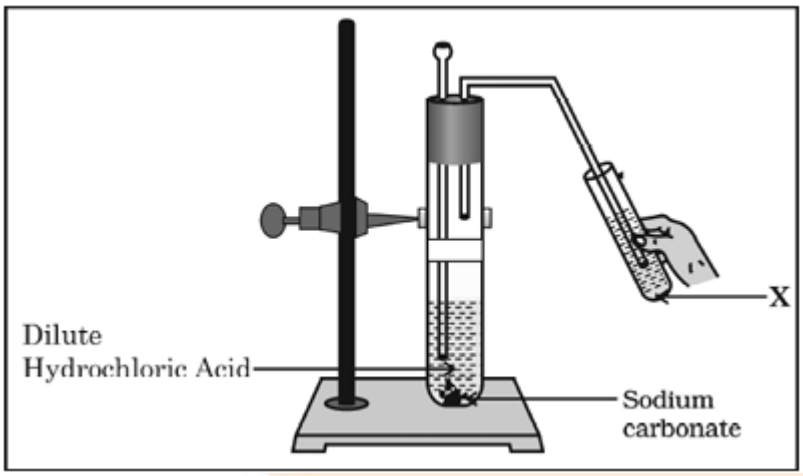
# CBSE Class 10 Science Question Paper 2023

## Set 1 (31/1/1) Solutions

### SECTION – A

**Select and write one most appropriate option out of the four options given for each of the questions 1 – 20:**

**1. In the experimental setup given below, it is observed that on passing the gas produced in the reaction in the solution 'X' the solution 'X' first turns milky and then colourless.** **1**




**The option that justifies the above stated observation is that 'X' is aqueous calcium hydroxide and:**

**(a) it turns milky due to carbon dioxide gas liberated in the reaction and after some time it becomes colourless due to formation of calcium carbonate.**

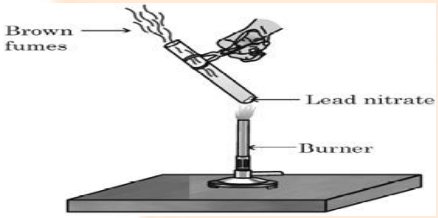
**(b) it turns milky due to formation of calcium carbonate and on passing excess of carbon dioxide it becomes colourless due to formation of calcium hydrogen carbonate which is soluble in water.**

**(c) it turns milky due to passing of carbon dioxide through it. It turns colourless as on further passing carbon dioxide, sodium hydrogen carbonate is formed which is soluble in water.**

**(d) the carbon dioxide liberated during the reaction turns lime water milky due to formation of calcium hydrogen carbonate and after some time it turns colourless due to formation of calcium carbonate which is soluble in water.**

**Answer: (**b) it turns milky due to formation of calcium carbonate and on passing excess of carbon dioxide it becomes colourless due to formation of calcium hydrogen carbonate which is soluble in water.

**2. The emission of brown fumes in the given experimental set-up is due to 1**




**(a) thermal decomposition of lead nitrate which produces brown fumes of nitrogen dioxide.**

**(b) thermal decomposition of lead nitrate which produces brown fumes of lead oxide.**

**(c) oxidation of lead nitrate forming lead oxide and nitrogen dioxide.**

**(d) oxidation of lead nitrate forming lead oxide and oxygen.**

**Answer:** (a) thermal decomposition of lead nitrate which produces brown fumes of nitrogen dioxide.

**3.** $MnO_2 + xHCl \rightarrow MnCl_2 + yH_2O + zCl_2$ **1**

**In order to balance the above chemical equation, the values of x, y and z respectively are :**

**(a) 6, 2, 2**

**(b) 4, 1, 2**

**(c) 4, 2, 1**

**(d) 2, 2, 1**

**Answer:** (c) 4, 2, 1

**4. The table below has information regarding pH and the nature (acidic/basic) of four different solutions. Which one of the options in the table is correct?** **1**

| Option | Solution | Colour of pH paper | Approximate pH value | Nature of solution |
|---|---|---|---|---|
| (a) | Lemon juice | Orange | 3 | Basic |
| (b) | Milk of magnesia | Blue | 10 | Basic |
| (c) | Gastric juice | Red | 6 | Acidic |
| (d) | Pure water | Yellow | 7 | Neutral |

**Answer:** (c) Gastric juice

**5. A metal 'X' is used in the thermite process. When X is burnt in air it gives an amphoteric oxide 'Y'. 'X' and 'Y' are respectively: 1**

**(a) Fe and $Fe_2O_3$**

**(b) Al and $Al_2O_3$**

**(c) Fe and $Fe_3O_4$**

**(d) Al and $Al_2O_4$**

**Answer:** (b) Al and $Al_2O_3$

**6. Select washing soda from the following: 1**

$(a) NaHCO_3$

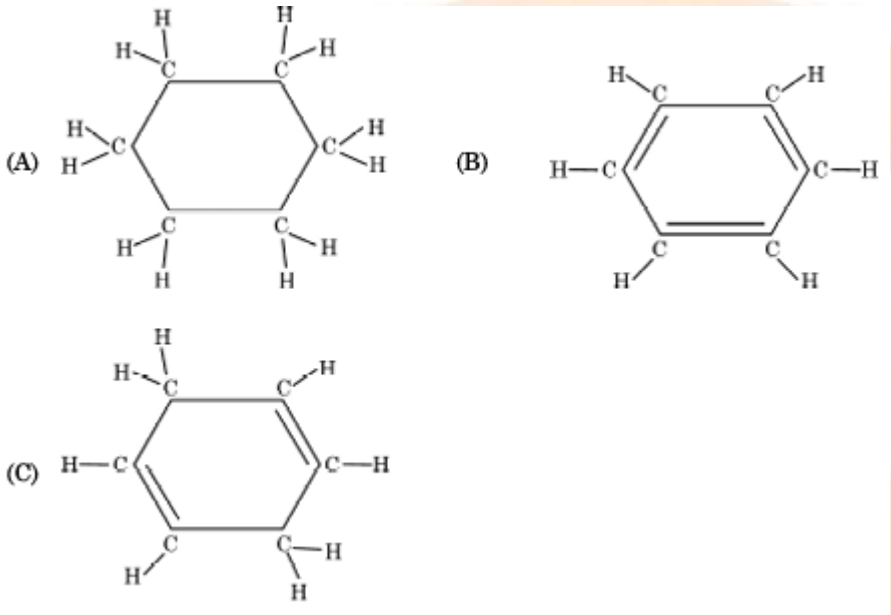
$(b) Na_2CO_3 \cdot 5H_2O$

(c) $Na_2CO_3 \cdot 10H_2O$

(d) NaOH

**Answer:** $(b) Na_2CO_3 \cdot 5H_2O$

**7. Consider the structures of the three cyclic carbon compounds A, B and C given below and select the correct option from the following: 1**




**(a) A and C are isomers of hexane and B is benzene.**

**(b) A is an isomer of hexane, B is benzene and C is an isomer of hexene.**

**(c) A is a saturated cyclic hydrocarbon and B and C are unsaturated cyclic hydrocarbons.**

**(d) A is cyclohexane and B and C are the isomers of benzene.**

**Answer:** (c) A is a saturated cyclic hydrocarbon and B and C are unsaturated cyclic hydrocarbons.

**8. An organism which breaks down the food material outside the body and then absorbs it is** **1**

**(a) a plant parasite, Cuscuta**

**(b) an animal parasite, Tapeworm**

**(c) a bacteria, Rhizobium**

**(d) a fungi, Rhizopus**

**Answer:** (d) a fungi, Rhizopus

**9. Consider the following statements about small intestine and select the one which is NOT correct:** **1**

**(a) The length of the small intestine in animals differs as it depends on the type of food they eat.**

**(b) The small intestine is the site of complete digestion of food.**

**(c) The small intestine receives secretions from liver and pancreas.**

**(d) The villi of the small intestine absorb water from the unabsorbed food before it gets removed from the body via the anus.**

**Answer:** (d) The villi of the small intestine absorb water from the unabsorbed food before it gets removed from the body via the anus.

**10. The statement that correctly describes the characteristic(s) of a gene is: 1**

**(a) In individuals of a given species, a specific gene is located on a particular chromosome.**

**(b) A gene is not the information source for making proteins in the cell.**

**(c) Each chromosome has only one gene located all along its length.**

**(d) All the inherited traits in human beings are not controlled by genes.**

**Answer:** (a) In individuals of a given species, a specific gene is located on a particular chromosome.

**11. Select from the following the correct statement about tropic movement in plants: 1**

**(a) It is due to stimulus of touch and temperature.**

**(b) It does not depend upon the direction of stimulus received.**

**(c) It is observed only in roots and not in stems.**

**(d) It is a growth related movement.**

**Answer:** (d) It is a growth related movement.

**12. Select the INCORRECT match (between the plant and its vegetative part) from the following: 1**

(a) Bryophyllum, leaf

(b) Potato, stem

(c) Money-plant, stem

(d) Rose, root

**Answer:** (d) Rose, root

**13. If four identical resistors, of resistance 8 ohm, are first connected in series so as to give an effective resistance Rs , and then connected in parallel so as to give an effective resistance Rp, then the ratio $\frac{R_s}{R_p}$ is 1**

(a) 32

(b) 2

(c) 0.5

(d) 16

**Answer:** (a) 32

**14. In domestic electric circuits the wiring with 15 A current rating is for the electric devices which have** **1**

**(a) higher power ratings such as geyser.**

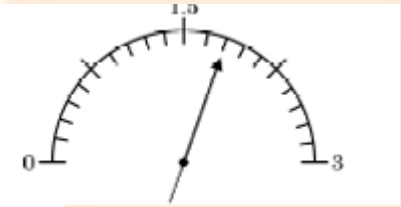
**(b) lower power ratings such as fan.**

**(c) metallic bodies and low power ratings.**

**(d) non-metallic bodies and low power ratings.**

**Answer:** (a) higher power ratings such as geyser.

**15. In the following diagram, the position of the needle is shown on the scale of a voltmeter. The least count of the voltmeter and the reading shown by it respectively are:** **1**




**(a) 0.15 V and 1.6 V**

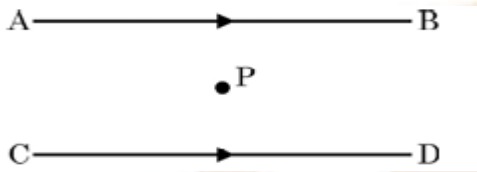
**(b) 0.05 V and 1.6 V**

**(c) 0.15 V and 1.8 V**

**(d) 0.05 V and 1.8 V**

**Answer:** (b) 0.05 V and 1.6 V

**16. The resultant magnetic field at point 'P' situated midway between two parallel wires (placed horizontally) each carrying a steady current I is** **1**




**(a) in the same direction as the current in the wires.**

**(b) in the vertically upward direction.**

**(c) zero.**

**(d) in the vertically downward direction.**

**Answer:** (c) zero

**Q. Nos. 17 to 20 are Assertion – Reason based questions.**

**These consist of two statements – Assertion (A) and Reason (R). Answer these questions selecting the appropriate option given below :**

**(a) Both (A) and (R) are true and (R) is the correct explanation of (A).**

**(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).**

**(c) (A) is true, but (R) is false.**

**(d) (A) is false, but (R) is true.**

**17. Assertion (A): The colour of aqueous solution of copper sulphate turns colourless when a piece of lead is added to it.**
**Reason (R): Lead is more reactive than copper, and hence displaces copper from its salt solution. 1**

**Answer:** (d) (A) is false, but (R) is true.

**18. Assertion (A): Genes inherited from the parents decide the sex of a child.**
**Reason (R): X chromosome in a male child is inherited from his father. 1**

**Answer:** (c) (A) is true, but (R) is false.

**19. Assertion (A): Blood clotting prevents excessive loss of blood.**
**Reason (R): Blood clotting is due to blood plasma and white blood cells present in the blood. 1**

**Answer:** (c) (A) is true, but (R) is false.

**20. Assertion (A): The strength of the magnetic field produced at the centre of a current carrying circular coil increases on increasing the number of turns in it.**
**Reason (R): The current in each circular turn has the same direction and the magnetic field due to each turn then just adds up. 1**

**Answer:** (a) Both (A) and (R) are true and (R) is the correct explanation of (A).

## SECTION–B

**Q. No. 21 to 26 are very short answer questions.**

**21. (a) (i) A compound 'X' which is prepared from gypsum has the property of hardening when mixed with proper quantity of water.**
**Identify 'X' and write its chemical formula. 2**

**Answer:** 'X' is Plaster of Paris. Its chemical formula is $CaSO_4 . \frac{1}{2} H_2O$.

**(ii) State the difference in chemical composition between baking soda and baking powder.**

**Answer:** Baking soda is sodium bicarbonate $NaHCO_3$, whereas baking powder is a mixture of sodium bicarbonate and an acid (usually cream of tartar) along with a drying agent (usually starch).

**OR**

**(b) Write balanced chemical equation for the reaction that occurs when : 2**

**(i) blue coloured copper sulphate crystals are heated**

**Answer:** $CuSO_4 \cdot 5H_2O \xrightarrow{heat} CuSO_4 + 5H_2O$

**(ii) Sodium hydrogen carbonate is heated during cooking.**

**Answer:** $2NaHCO_3 \xrightarrow{heat} Na_2CO_3 + H_2O + CO_2$

**22. (a) Write the role of insulin in regulating blood sugar levels in human body. Mention the disease caused due to it. 2**

**Answer:** Insulin helps in regulating blood sugar levels by facilitating the uptake of glucose into cells, thereby lowering blood sugar levels. The disease caused due to lack of insulin or insulin resistance is diabetes.

**(b) How is the timing and the amount of release of insulin in the blood regulated?**

**Answer:** The release of insulin is regulated by the blood glucose levels. When blood glucose levels rise after eating, the pancreas releases insulin to help cells absorb glucose. The amount of insulin released is proportional to the blood glucose concentration.

**23. (a) Name the type of blood (oxygenated / deoxygenated) transported by each of the following mentioning the path (i.e. from one organ (which place) to another (which place)). 2**

**(i) Vena cava**

**Answer:** Vena cava carries deoxygenated blood from the body to the right atrium of the heart.

**(ii) Pulmonary artery 2**

**Answer:** Pulmonary artery carries deoxygenated blood from the right ventricle of the heart to the lungs.

**OR**

**(b) With the help of a schematic flow chart, show the breakdown of glucose in a cell to provide energy in: 2**

**(i) in the presence of oxygen**

**Answer:**
Glucose→Glycolysis→Pyruvate→Krebs cycle→Electron transport chain→ATP (Energy)

**(ii) in the absence of oxygen.**

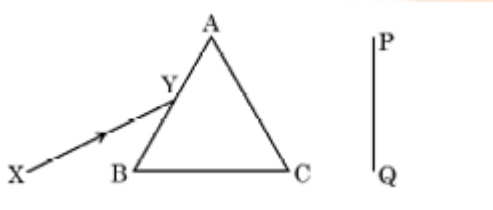
**Answer:**
Glucose→Glycolysis→Pyruvate→Fermentation→Lactic acid or Ethanol→Less ATP (Energy)

**24. Name the part of the human excretory system where nephrons are found. Write the structure and function of nephrons. 2**

**Answer:** Nephrons are found in the kidneys.

- **Structure:** Each nephron consists of a glomerulus, Bowman's capsule, proximal convoluted tubule, loop of Henle, distal convoluted tubule, and collecting duct.
- **Function:** Nephrons filter blood to remove waste products and excess substances, forming urine which is excreted from the body.

**25. (a) A narrow beam XY of white light is passing through a glass prism ABC as shown in the diagram: 2**




**Trace it on your answer sheet and show the path of the emergent beam as observed on the screen PQ.**
**Name the phenomenon observed and state its cause.**

**Answer:** The phenomenon observed is dispersion of light. This occurs because different wavelengths of light refract at different angles when passing through the prism, causing the white light to split into its constituent colors.

**OR**

**(b) It is observed that the power of an eye to see nearby objects as well as far off objects diminishes with age. 2**

**(i) Give reason for the above statement.**

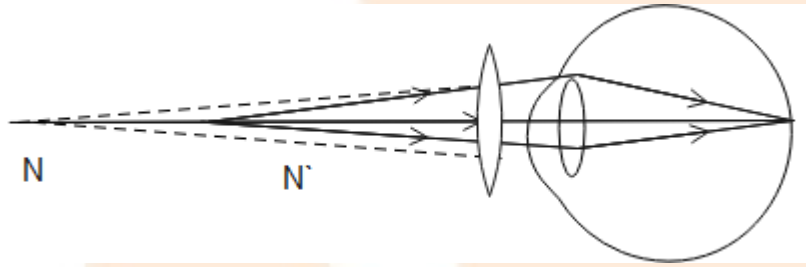
**Answer:** This condition is due to the loss of elasticity of the eye lens, which affects its ability to change shape and focus on objects at different distances.

**(ii) Name the defect that is likely to arise in the eyes in such a condition.**

**Answer:** The defect is presbyopia.

**(iii) Draw a labelled ray diagram to show the type of corrective lens used for restoring the vision of such an eye.**

**Answer:**




**26. How do harmful chemicals get accumulated progressively at each trophic level in a food chain? 2**

**Answer:** Harmful chemicals, such as pesticides, enter the food chain at the producer level. As each trophic level consumes the previous one, these chemicals accumulate in higher concentrations due to biomagnification. This leads to progressively higher concentrations of these chemicals in the tissues of organisms at higher trophic levels, causing various health issues.

## SECTION–C

**Q. No. 27 to 33 are short answer questions.**

**27. (a) Identify the reducing agent in the following reactions: 3**

**(i)** $NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$

**Answer:** The reducing agent is $NH_3$ (ammonia).

**(ii)** $H_2O + F_2 \rightarrow HF + HOF$

**Answer:** The reducing agent is $H_2O$

**(iii)** $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$

**Answer:** The reducing agent is $CO$ (carbon monoxide).

**(iv)** $H_2 + O_2 \rightarrow 2H_2O$

**Answer:** The reducing agent is $H_2$ (hydrogen).

**(b) Define a redox reaction in terms of gain or loss of oxygen.**

**Answer:** A redox reaction is a chemical reaction in which there is a transfer of electrons between two species. It involves the gain of oxygen (oxidation) or the loss of oxygen (reduction) by the reactants.

**28. (a) Suggest one remedial measure each to counteract the change in pH in human beings in the following cases: 3**

**(i) Production of too much acid in stomach during indigestion**

**Answer:** Take an antacid such as magnesium hydroxide or sodium bicarbonate to neutralize the excess stomach acid.

**(ii) Stung by a honey bee / nettle leaves**

**Answer:** Apply a paste of baking soda and water to neutralize the acidic venom from a bee sting. Apply a paste of vinegar or lemon juice to neutralize the alkaline effect of nettle leaves sting.

**(b) Fresh milk has a pH of 6. When it changes into curd will its pH increase or decrease? Why?**

**Answer:** The pH will decrease because the fermentation process produces lactic acid, which lowers the pH, making the curd more acidic.

**29. (a) (i) State the role of ATP in cellular respiration.** **3**

**Answer:** ATP (adenosine triphosphate) serves as the main energy currency of the cell, providing energy for various cellular processes by releasing energy stored in its high-energy phosphate bonds during cellular respiration.

**(ii) What ensures sufficient exchange of gases in plants?**

**Answer:** The presence of stomata on the leaves ensures sufficient exchange of gases (oxygen and carbon dioxide) in plants.

**(iii) State the conditions on which the direction of diffusion of gases in plants depend upon.**

**Answer:** The direction of diffusion of gases in plants depends on the concentration gradient of the gases and the presence of stomata that facilitate the gas exchange.

**OR**

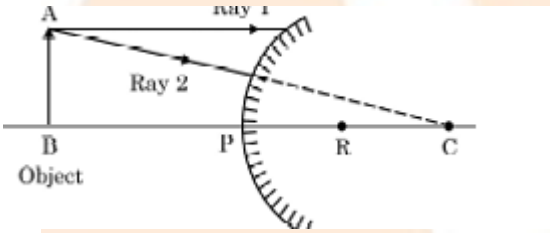
**(b) (i) What is the internal energy reserve in plants and animals?** **3**

**Answer:** In plants, the internal energy reserve is starch, while in animals it is glycogen.

**(ii) How desert plants perform photosynthesis if their stomata remain closed during the day?**

**Answer:** Desert plants perform photosynthesis by using the CAM (Crassulacean Acid Metabolism) pathway, which allows them to take in CO2_22 at night and store it as malic acid. During the day, they close their stomata to conserve water and use the stored CO2_22 for photosynthesis.

**30. (a) Complete the following ray diagram to show the formation of image: 3**




**Answer:** (Diagram completion: Ray 1 and Ray 2 should be drawn to meet at point R, forming the image)

**(b) Mention the nature, position and size of the image formed in the case**.

**Answer:** The image formed is real, inverted, and diminished in size.

**(c)** State the type of mirror that has been used in this case using the characteristics of the image formed.

**Answer:** The type of mirror used is a concave mirror because it forms a real, inverted, and diminished image when the object is placed beyond the center of curvature.

**31. Give reasons for the following:** **3**

**(a) Danger signals installed at airports and at the top of tall buildings are of red colour.**

**Answer:** Red colour has the longest wavelength among all visible colours and is scattered the least by atmospheric particles. This makes it visible from a greater distance, ensuring that danger signals can be seen clearly.

**(b) The sky appears dark to the passengers flying at very high altitudes.**

**Answer:** At high altitudes, the atmosphere is much thinner and contains fewer particles to scatter sunlight. As a result, there is less scattering of light, and the sky appears dark.

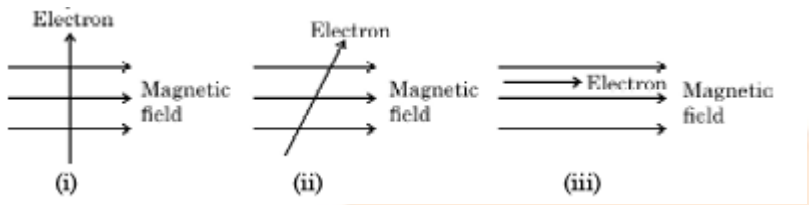
**(c) The path of a beam of light passing through a colloidal solution is visible.**

**Answer:** The path of light is visible due to the Tyndall effect, where light is scattered by the colloidal particles in the solution, making the path of the beam visible.

**32. (a) (i)** State the rule used to find the force acting on a current-carrying conductor placed in a magnetic field.

**Answer:** The rule used is Fleming's Left Hand Rule, which states that if the thumb, forefinger, and middle finger of the left hand are extended perpendicular to each other, with the forefinger pointing in the direction of the magnetic field and the middle finger in the direction of current, then the thumb will point in the direction of the force acting on the conductor.

**(ii)** Given below are three diagrams showing entry of an electron in a magnetic field. Identify the case in which the force will be (1) maximum and (2) minimum respectively. Give reason for your answer.




**Answer:**

1. Maximum force: (ii) - The force is maximum when the direction of the electron's velocity is perpendicular to the magnetic field.

2. Minimum force: (i) - The force is minimum (zero) when the direction of the electron's velocity is parallel to the magnetic field.

**OR**

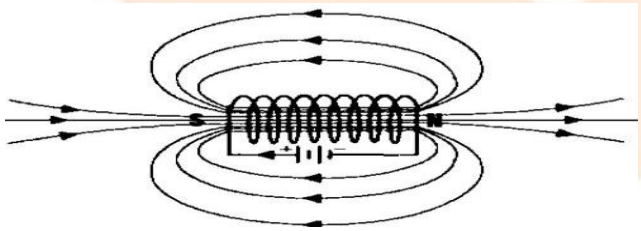
**(b) (i) Draw the pattern of magnetic field lines of: 3**

1. **A current-carrying solenoid**

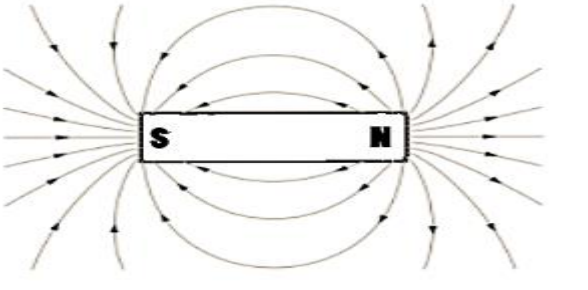
2. **A bar magnet**

**Answer:**

1. Current-carrying solenoid: Magnetic field lines are parallel inside the solenoid, indicating a uniform magnetic field, and curve outside the solenoid, resembling the field lines of a bar magnet.




2. Bar magnet: Magnetic field lines emerge from the north pole and enter the south pole, forming closed loops.




**(ii) List two distinguishing features between the two fields.**

**Answer:**

1. A current-carrying solenoid creates a uniform magnetic field inside it, while a bar magnet has a non-uniform magnetic field.

2. The magnetic field of a solenoid can be turned on or off by controlling the current, while the magnetic field of a bar magnet is permanent.

**33. (a) (i) Why does a kitchen garden called an artificial ecosystem while a forest is considered to be a natural ecosystem? 3**

**Answer:** A kitchen garden is called an artificial ecosystem because it is created and maintained by humans, involving planned planting and care. A forest is a natural ecosystem because it develops naturally without human intervention.

**(ii) While designing an artificial ecosystem at home, write any two things to be kept in mind to convert it into a self-sustaining system. Give reason to justify your answer.**

**Answer:**

1. Ensure a balance of producers, consumers, and decomposers to maintain nutrient cycling.

2. Provide sufficient water and sunlight to support the growth of plants and maintain ecological balance.

**OR**

**(b) (i) Construct a food chain of four trophic levels comprising the following: Hawk, snake, plants, rat. 3**

**Answer:** Plants (producers) → Rat (primary consumer) → Snake (secondary consumer) → Hawk (tertiary consumer)

**(ii) 20,000 J of energy was transferred by the producers to the organisms of second trophic level. Calculate the amount of energy which will be transferred by organisms of the third trophic level to the organisms of the fourth trophic level.**

**Answer:** According to the 10% law, only 10% of the energy is transferred to the next trophic level.

Energy at third trophic level = 10% of 20,000 J = 2,000 J

Energy transferred to the fourth trophic level = 10% of 2,000 J = 200 J

## SECTION–D

**Q. No. 34 to 36 are long answer questions.**

**34. (a) A saturated organic compound 'A' belongs to the homologous series of alcohols. On heating 'A' with concentrated sulphuric acid at 443 K, it forms an unsaturated compound 'B' with molecular mass 28 u. The compound 'B' on addition of one mole of hydrogen in the presence of Nickel, changes to a saturated hydrocarbon 'C'. 5**

**(i) Identify A, B and C.**

**Answer:**

- A is Ethanol $(C_2H_5OH)$

- B is Ethene $(C_2H_4)$
- C is Ethane $(C_2H_6)$

**(ii) Write the chemical equations showing the conversion of A into B.**

**Answer:** $C_2H_5OH \xrightarrow{\text{Conc. } H_2SO_4 \text{ at } 443\text{ K}} C_2H_4 + H_2O$

**(iii) What happens when compound C undergoes combustion?**

**Answer:** When Ethane (C) undergoes combustion, it reacts with oxygen to produce carbon dioxide, water, and releases energy. 2C2H6+7O2→4CO2+6H2O+Energy

$2C_2H_6 + 7O_2 \rightarrow 4CO_2 + 6H_2O + \text{Energy}$

**(iv) State one industrial application of hydrogenation reaction.**

**Answer:** One industrial application of hydrogenation is the conversion of vegetable oils into solid fats (margarine) using hydrogen in the presence of a nickel catalyst.

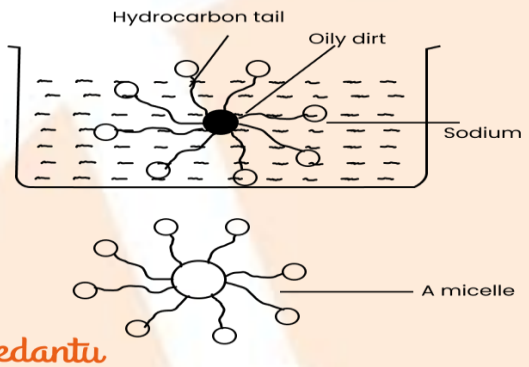
**(v) Name the products formed when compound A reacts with sodium.**

**Answer:** When Ethanol (A) reacts with sodium, the products formed are sodium ethoxide and hydrogen gas.

$2C_2H_5OH + 2Na \rightarrow 2C_2H_5ONa + H_2$

**OR**

**(b) (i) With the help of diagram, show the formation of micelles, when soap is applied on oily dirt. 5**




**Answer:**

(Description: A diagram showing soap molecules forming micelles around oil droplets, with hydrophobic tails inwards and hydrophilic heads outwards.)

**(ii)** Take two test tubes X and Y with 10 mL of hard water in each. In test tube 'X', add few drops of soap solution and in test tube 'Y' add a few drops of detergent solution. Shake both the test tubes for the same period.

1. **In which test tube the formation of foam will be more? Why?**

**Answer:** The formation of foam will be more in test tube 'Y' with the detergent solution because detergents do not react with the calcium and magnesium ions in hard water, whereas soap forms scum with these ions, reducing foam formation.

2. **In which test tube a curdy solid is formed? Why?**

**Answer:** A curdy solid is formed in test tube 'X' with the soap solution because soap reacts with the calcium and magnesium ions present in hard water to form an insoluble precipitate called scum.

**35. (a) Name the parts of a bisexual flower that are not directly involved in reproduction. 5**

**Answer:** The parts of a bisexual flower that are not directly involved in reproduction include the petals, sepals, and bracts.

**(b) Differentiate between self-pollination and cross-pollination. List any two significance of pollination.**

**Answer:**

- **Self-pollination:** The transfer of pollen from the anther to the stigma of the same flower or another flower on the same plant.
- **Cross-pollination:** The transfer of pollen from the anther of one flower to the stigma of a flower on a different plant of the same species.

**Significance of pollination:**

1. Pollination leads to fertilization, resulting in the formation of seeds and fruits.
2. Cross-pollination increases genetic diversity, which enhances the adaptability and survival of plant species.

**(c) What is the fate of ovules and ovary after fertilization in a flower?**

**Answer:** After fertilization, the ovules develop into seeds, and the ovary develops into a fruit.

**36. (a) An electric iron consumes energy at a rate of 880 W when heating is at the maximum rate and 330 W when the heating is at the minimum. If the source voltage is 220 V, calculate the current and resistance in each case. 5**

**Answer:**

- **At maximum heating:**

  Power=880 W

  Voltage=220 V

  Current= $\frac{\text{Power}}{\text{Voltage}} = \frac{880}{220} = 4\text{ A}$

  $\text{Resistance} = \frac{\text{Voltage}}{\text{Current}} = \frac{220}{4} = 55\text{ ohms}$

- **At minimum heating:**

  Power=330 W

  Voltage=220 V

  $\text{Current} = \frac{\text{Power}}{\text{Voltage}} = \frac{330}{220} = 1.5\text{ A}$

  $\text{Resistance} = \frac{\text{Voltage}}{\text{Current}} = \frac{220}{1.5} \approx 146.67\text{ ohms}$

  Resistance=CurrentVoltage=1.5220≈146.67 ohms

**(b) What is the heating effect of electric current?**

**Answer:** The heating effect of electric current is the phenomenon where electrical energy is converted into heat energy when an electric current passes through a conductor. This is due to the resistance offered by the conductor to the flow of current.

**(c)** Find an expression for the amount of heat produced when a current passes through a resistor for some time.

**Answer:** The amount of heat (H) produced when a current (I) passes through a resistor (R) for a time (t) is given by Joule's law of heating:

$H = I^2 Rt$

## SECTION–E

**Q. No. 37 to 39 are case-based/data-based questions with 2 to 3 short sub-parts. Internal choice is provided in one of these sub-parts.**

**37. Almost all metals combine with oxygen to form metal oxides. Metal oxides are generally basic in nature. But some metal oxides show both basic as well as acidic behaviour. Different metals show different reactivities towards oxygen. Some react vigorously while some do not react at all.**

**(a) What happens when copper is heated in air? (Give the equation of the reaction involved.) 1**

**Answer:** When copper is heated in air, it reacts with oxygen to form copper(II) oxide (CuO), a black solid.

$$Cu + O_2 \rightarrow 2CuO$$

**(b) Why are some metal oxides categorized as amphoteric? Give one example.** **1**

**Answer:** Some metal oxides are categorized as amphoteric because they can react with both acids and bases to form salts and water. An example of an amphoteric oxide is aluminum oxide $(Al_2O_3)$.

**(c) Complete the following equations:** **2**

1. $Na_2O + H_2O \rightarrow$

   **Answer:** $Na_2O + H_2O \rightarrow 2NaOH$

2. $Al_2O_3 + 2NaOH \rightarrow$

   **Answer:** $Al_2O_3 + 2NaOH + 3H_2O \rightarrow 2NaAl(OH)_4$

**(d) On burning Sulphur in oxygen a colourless gas is produced.** **2**

**(i) Write chemical equation for the reaction.**

**Answer:** $\mathbf{S + O_2 \rightarrow SO_2}$

**(ii) Name the gas formed.**

**Answer:** The gas formed is Sulphur dioxide $(SO_2)$.

**(iii) State the nature of the gas.**

**Answer:** Sulphur dioxide is an acidic gas.

**(iv) What will be the action of this on a dry litmus paper?**

**Answer:** Sulphur dioxide will turn blue litmus paper red because it is an acidic gas.

**38. In order to trace the inheritance of traits Mendel crossed pea plants having one contrasting character or a pair of contrasting characters. When he crossed pea plants having round and yellow seeds with pea plants having wrinkled and green seeds, he observed that no plants with wrinkled and green seeds were obtained in the F, generation. When the F1 generation pea plants were cross-bred by self-pollination, the F2 generation had seeds with different combinations of shape and colour also.**

**(a) Write any two pairs of contrasting characteristics of pea plant used by Mendel other than those mentioned above. 1**

**Answer:**

1. Tall and dwarf plants
2. Green and yellow pods

**(b) Differentiate between dominant and recessive traits. 1**

**Answer:**

- Dominant traits are those that express themselves in the phenotype of the organism even if only one copy of the allele is present. For example, in Mendel's pea plants, round seeds and yellow seeds are dominant traits.
- Recessive traits are expressed in the phenotype only when two copies of the recessive allele are present. For example, wrinkled seeds and green seeds are recessive traits.

**(c) State the ratio of the combinations observed in the seeds of F2 generation (in the above case). What do you interpret from this result? 2**

**Answer:** The ratio of the combinations observed in the $F_2$ generation is 9:3:3:1, where:

- 9 are round and yellow
- 3 are round and green
- 3 are wrinkled and yellow
- 1 is wrinkled and green

Interpretation: This result indicates that the traits are independently assorted, showing the principle of independent assortment. The ratio also reflects the dominance and recessiveness of the traits.

**OR**

**(c) Given below is a cross between a pure violet flowered pea plant (V) and a pure white flowered pea plant (v). Diagrammatically explain what type of progeny is obtained in F, generation and F, generation: 2**

**Pure violet flowered plant x Pure white flowered plant.**

(VV) (vv)

**Answer:**

1. $F_1$ **Generation:**

When a pure violet-flowered plant (VV) is crossed with a pure white-flowered plant (vv), all the offspring in the $F_1$ generation will have the genotype Vv and display the violet flower phenotype.

**Diagram:**

Parent (VV)×Parent (vv)

F1:All Vv (violet flowers)

2. **F2_22 Generation:**

When F1_11 plants (Vv) are self-pollinated, the resulting $F_2$ generation will have a phenotypic ratio of 3:1, where 3/4 will be violet-flowered and 1/4 will be white-flowered.

**Diagram:**

F1(Vv)×F1(Vv)

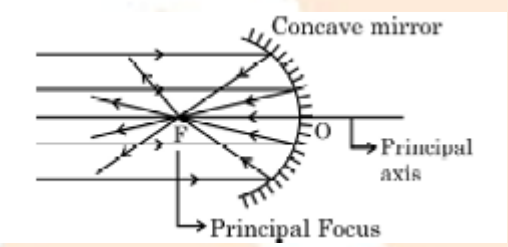
**Punnett Square:**

| | $V$ | $v$ |
|---|---|---|
| $V$ | $VV$ | $Vv$ |
| $v$ | $Vv$ | $vv$ |

**F2_22 Ratio:**

- Violet flowers (VV and Vv): 75%
- White flowers (vv): 25%

**39. Hold a concave mirror in your hand and direct its reflecting surface towards the sun. Direct the light reflected by the mirror on to a white card-board held close to the mirror. Move the card-board back and forth gradually until you find a bright, sharp spot of light on the board. This spot of light is the image of the sun on the sheet of paper; which is also termed as "Principal Focus" of the concave mirror.**




**(a) List two applications of concave mirror.** **1**

**Answer:**

1. Shaving Mirrors: Concave mirrors are used in shaving mirrors to provide a magnified image of the face.
2. Reflecting Telescopes: Concave mirrors are used in reflecting telescopes to gather and focus light from distant celestial objects.

**(b) If the distance between the mirror and the principal focus is 15 cm, find the radius of curvature of the mirror.** **1**

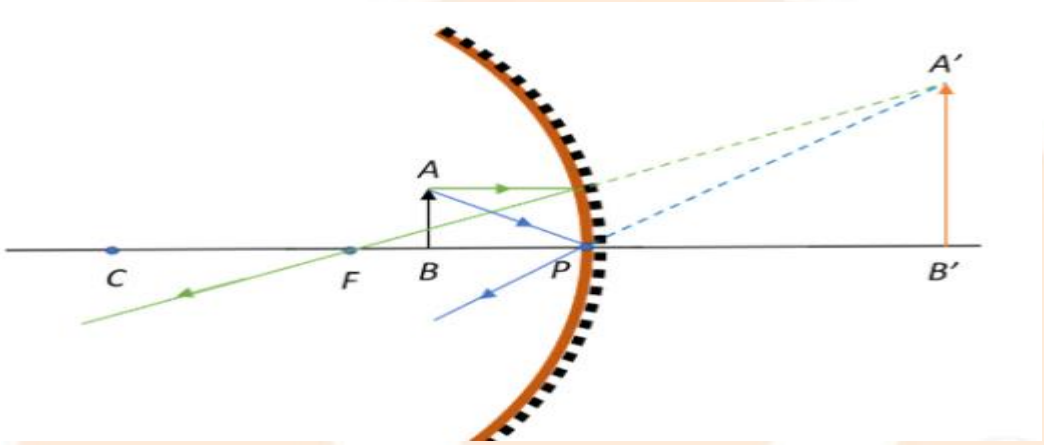
**Answer:** The distance between the mirror and the principal focus is the focal length (f) of the mirror. The radius of curvature (R) is related to the focal length by the formula:

$R=2f$

Given $f=15$ cm

$R=2\times15=30$ cm

**(c) Draw a ray diagram to show the type of image formed when an object is placed between pole and focus of a concave mirror. 2**




Ray Diagram:

1. Object: Place the object between the pole (P) and the principal focus (F) of the concave mirror.
2. Incident Ray 1: Draw a ray parallel to the principal axis, which reflects through the principal focus (F).
3. Incident Ray 2: Draw a ray passing through the focus (F) and reflecting parallel to the principal axis.
4. Reflected Ray: Extend the reflected rays backward to meet at a point on the same side of the mirror as the object.

**The image formed will be:**

- Location: Beyond the principal focus (F) on the same side of the mirror as the object.
- Nature: Virtual, erect, and magnified.

**OR**

**(c) An object 10 cm in size is placed at 100 cm in front of a concave mirror. If its image is formed at the same point where the object is located, find: 2**

**(i) focal length of the mirror, and**

If the image is formed at the same point as the object, the image distance (v) is equal to the object distance (u). So, v=−u (since it's a concave mirror and distances are taken as negative).

Given:

- Object distance, u=−100 cm
- Image distance, v=−100 cm

Using the mirror formula: $\frac{1}{f}=\frac{1}{u}+\frac{1}{v}$

$$\frac{1}{f}=\frac{1}{-100}+\frac{1}{-100}$$

$$\frac{1}{f}=\frac{-2}{100}$$

f=−50 cm

**(ii) magnification of the image formed with sign as per Cartesian sign convention.**

Answer:

Magnification (M) is given by:

M=$\frac{v}{u}$

$$M = \frac{-100}{-100} = 1$$

**Sign convention:**

- Since the magnification is positive, the image is erect and the same size as the object.